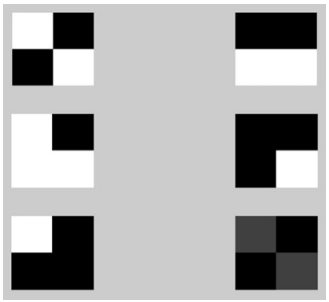
Lab-1

1. Write a Python program using OpenCV to display an image and its negative for both binary and grayscale formats.
2. Write a Python program using OpenCV to read a color image and display its individual color components (Red, Green, and Blue).
3. Write a program to resize an image to different resolutions (e.g., 256 × 256 and 512 × 512) and compare the results.
4. Write a program to rotate an image by **45°**, **90°**, and **180°**, and display the rotated images.
5. The following figure displays six grayscale images generated from matrix operations involving binary matrices.



Write a Python program using **NumPy** and **Matplotlib** to reproduce the figure as shown.

Hints:

a. Create two 40×40 matrices: one of ones (a), another of zeros (b)
b. Construct:
   a. c = [[a, b], [b, a]]
   b. d = [[b, b], [a, a]] using np. block ()

Perform:

a. A = 10 * (c + d)
b. M = c * d
c. S = c - d
d. D = c / 4

Display:

a. Show c, d, A, M, S, and D using matplotlib. pyplot. imshow() in a **3×2 subplot**

6. Write a Python program using NumPy to find the 4-neighbors, 8-neighbors, and diagonal neighbors of a given pixel in a 5×5 matrix.
   Instructions:
   - Use a 5×5 matrix (e.g., np.random.randint() or np.arange(). reshape () or magic square equivalent).
   - Take input from the user for a pixel location (row, col).

   Display:
   - The selected pixel value.
   - 4-neighbors (top, bottom, left, right)
   - 8-neighbors (all surrounding pixels)
   - Diagonal neighbors (top-left, top-right, bottom-left, bottom-right)